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ARTICLE

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Extended kinetic lattice grand canonical Monte Carlo simulation method for transport of multicomponent ion mixtures through a model nanopore system

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Abstract

An extended version of the original kinetic lattice grand canonical Monte Carlo simulation method combined with mean field theory (KLGCMC/MF) (*J. Chem. Phys.* **2007**, 127, 024706) is presented for the study of transport of multicomponent ion mixtures through a model nanopore. Comparison of the extended KLGCMC/MF (eKLGCMC/MF) simulation results with Poisson–Nernst–Planck (PNP) calculations is also made to confirm the validity of the extended simulation approach. Unlike the original version of KLGCMC/MF simulation method that treats only a binary ionic solution with one cation and one anion species, this extended version can deal with a system that includes ternary ion mixtures. A diffusion probability algorithm is also added to the extended version of the simulation method to describe the inhomogeneous diffusivity of ions that is often observed in the ion permeation through nanopores. Both Legendre and Chebyshev polynomials of the second kind were tested as a basis set for the basis set expansion (BSE) method with which to calculate the reaction field energy in the eKLGCMC/MF simulation. It turned out that the Legendre polynomials perform better than the Chebyshev polynomials, and as a result, the Legendre polynomials were implemented in the current version of eKLGCMC/MF simulation algorithm. The presented eKLGCMC/MF simulation method with new features finds its potential applications in nanopore systems where the correlation between ion species with the same sign of charges plays a key role such as oscillating ion currents or anomalous mole fraction effects.

KEYWORDS
Grand canonical Monte Carlo simulation, Kinetic lattice simulation, Ion channel, Ion current, Nanopore

INTRODUCTION

Nanopores are narrow holes of nanometer size that are created by pore-forming proteins such as ion channels or synthetic materials like carbon nanotubes. There has been a great deal of interest in ion transport through nanopores due to their biological and nanotechnological implications.[1–3] The spatial restriction of nanopores renders interactions and correlations between permeating ions through nanopores much stronger than in the bulk. Such strong correlations between different ion species inside a narrow nanopore lead to many intriguing phenomena. For example, the anomalous mole fraction effect (AMFE) is a well-known phenomenon observed in multiion single file ion channels where strong correlations between two different ion species inside the small selectivity filter of an ion channel lead to unexpected nonlinear conductance curves.[4,5] It has also been reported that ion current oscillations can be generated by blocking proton currents through a single-walled carbon nanotube (SWNT) with the cations partitioning into the nanotube, demonstrating a strong correlation effect of the two different ion species inside the SWNT.[6,7] Stable ion current fluctuations in a

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conically shaped nanopore are another example that shows the correlation effect between different ion species. When the pore is exposed to weakly soluble salts which periodically block ion currents by repeatedly precipitating and dissolving at the small mouth of the pore.[8]

Kinetic (or dynamic) Monte Carlo (KMC or DMC) and Brownian dynamics (BD) simulations describe ion motions explicitly and can take accurate account of the ion correlation and reaction field effects, which are mostly ignored in the Poisson–Nernst–Planck (PNP) mean-field theory.[9,10] In addition, the static and dynamic properties of ions permeating through narrow nanopores can be obtained by the KMC and BD simulation approaches with reasonable computational times compared to what can be achieved by atomistic molecular dynamics (MD) simulations.[11–13] The long jump exception (LJE) problem due to unphysical ion configurations is often encountered in BD simulations for ion transport through a nanopore.[14] Unlike force-based BD simulations, the KMC simulation method is an energy-based method, and consequently, the LJE problem can be effectively avoided in the KMC method by using the Metropolis algorithm for ion moves.[15–20]

Taking advantages from the KMC method over the other methods, Hwang and coworkers proposed a kinetic lattice grand canonical Monte Carlo combined with the mean-field (KLGCMC/MF) simulation method for the study of ion transport through a model ion channel.[18] In this method, the core region of the system, including a model nanopore, is described using the kinetic lattice grand canonical Monte Carlo (KLGCMC) simulation algorithm while the rest of the system is treated with the Poisson–Boltzmann (PB) mean-field theory. The reaction field effect, which occurs when an ion approaches a dielecric boundary such as the membrane/water interface, is treated by the basis-set expansion (BSE) method in the simulation.[21] To acquire steady-state properties of the nanopore system, the local control volume technique is implemented in the simulation method where ion creations and deletions occur only in local control regions (LCRs) far from the core region.[22–25] Because the KLGCMC/MF simulation method is a lattice-based (or grid-based) algorithm, the simulation code can be easily parallelized to speed-up performance for a larger system.

In this study, we present an extension of the KLGCMC/MF (eKLGCMC/MF) simulation method, enabling us to explore the intriguing phenomena observed in nanopore systems with multicomponent ion mixtures.[4–8] While the original KLGCMC/MF method describes the transport of solely binary ion mixtures through a nanopore, the extended version can deal with the permeation of more than two sorts of ion species across a nanopore. The inhomogeneous diffusivity of ions is observed in the translocation of ions inside a small nanopore as well as among different kinds of ions in bulk water.[26–30] Unlike the original KLGCMC/MF simulation method that is unable to treat inhomogeneous ion diffusion, the extended version can handle the inhomogeneous diffusivity of ions by using a diffusion probability algorithm that was proposed for a lattice-based KMC simulation method.[31] In the original KLGCMC/MF simulation, Legendre polynomials were used as a basis set for the BSE method.[18] In this work, we test other functions such as Chebyshev polynomials of the second kind as well as Legendre functions to find a better set of functions as a basis set for the BSE algorithm. We compare the results from eKLGCMC/MF simulations with the PNP calculations to validate the extended version of the simulation method. This article is organized as follows. In the next section, the eKLGCMC/MF simulation method is proposed and a model nanopore system is introduced. In Section "Results and Discussion," the simulation outcomes for ion current–voltage (I-V) curves, ion concentration and electrostatic potential profiles in the

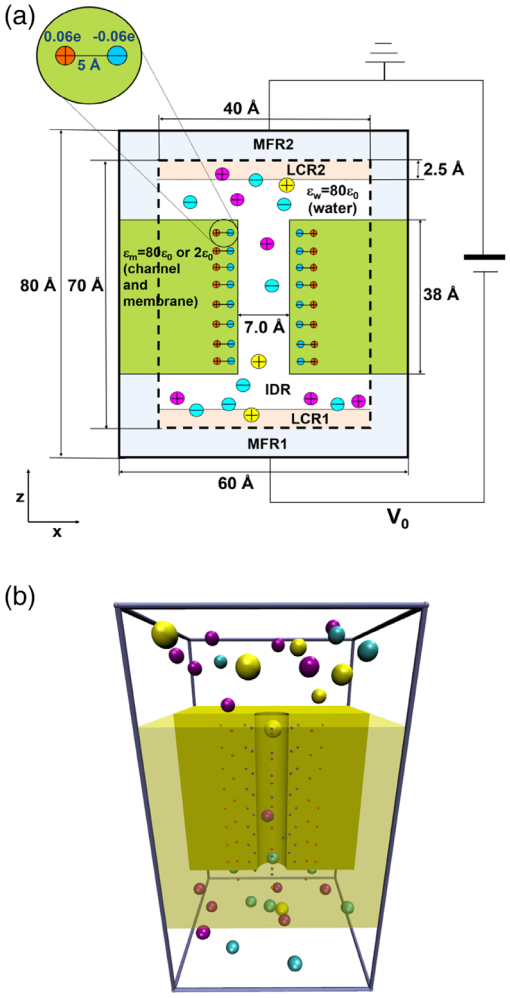


FIGURE 1 (a) Schematic representation of a model ion channel system and (b) snapshot from a eKLGCMC/MF simulation trajectory. In (a) the entire system is composed of two mean field regions (MFR1 and 2) and one Monte Carlo region (MCR). The MCR is then divided into two local control regions (LCR1 and 2) and one ion diffusion region (IDR). LCR1 and 2 are located at the bottom (-35 Å $\leq z \leq -32.5$ Å) and the top (32.5 Å $\leq z \leq$ 35 Å), respectively. In (b) only the MCR is represented along with the explicit ions, model ion channel and the fixed dipole moments embedded inside the implicit model membrane. The violet, yellow, and cyan spheres represent potassium and sodium cations and chloride anions, respectively, and the positive and negative ends of the fixed dipole moments are colored in red and blue, respectively

model system are compared with PNP calculations for several cases where interionic interactions and dielectric boundaries are absent or present. Applications as well as limitations and further improvements in the eKLGCMC/MF simulation method are addressed in the final section.

COMPUTATIONAL METHODS

eKLGCMC/MF simulation algorithm with the local control volume method

Since details of the original KLGCMC/MF simulation method are described in Ref. 18, we will only briefly address the new features in the extended version of the method. As shown in Figure 1a, a system consisting of a nanopore between two reservoirs is divided into three regions: mean-fields region 1 and 2 (MFR1 and MFR2) and the Monte Carlo region (MCR). The MFR1 and MFR2 regions are located far from the model nanopore and ions in the MFRs are described implicitly by the Poisson–Boltzmann (PB) equation. On the other hand, the MCR includes a nanopore system as shown in Figure 1b and ions in the MCR are treated explicitly by the kinetic grand canonical Monte Carlo method. The MCR is further divided into three regions: two local control regions (LCR1 and LCR2) where ions can be inserted/deleted to define a grand canonical ensemble, and one ion diffusion region (IDR) where insertion/deletion is not allowed.[25] The LCR1 and LCR2 are located at the bottom and top ends of the MCR, respectively, and the IDR is located to include the pore volume between the two LC regions (Figure 1a,b).

Ion creation is achieved by randomly selecting a lattice site in one of the two LCRs and attempting to create an ion i from multiple ion species at the selected lattice site. A creation attempt for that ion in the LCRs is accepted or rejected by generating and comparing a random number with a probability of $P_{N_i^{\mathrm{LCR1(or2)}} \to N_i^{\mathrm{LCR1(or2)}}+1}$:

$$P_{N_i^{\mathrm{LCR1(or2)}} \to N_i^{\mathrm{LCR1(or2)}}+1} = \begin{cases} \min\left\{1, \dfrac{\rho^{\mathrm{LCR1}}}{N_i^{\mathrm{LCR1}}+1}\exp\left[B_i^{\mathrm{LCR1}} - \beta\left(U_{N_i}{}^{\mathrm{LCR1}} + 1 - U_{N_i}{}^{\mathrm{LCR1}} - q_i^{\mathrm{cre}}V_0\right)\right]\right\}, (\mathrm{LCR1}), \\ \min\left\{1, \dfrac{\rho^{\mathrm{LCR1}}}{N_i^{\mathrm{LCR2}}+1}\exp\left[B_i^{\mathrm{LCR2}} - \beta\left(U_{N_i}{}^{\mathrm{LCR2}} + 1 - U_{N_i}{}^{\mathrm{LCR2}}\right)\right], \quad (\mathrm{LCR2}). \end{cases} \tag{1}$$

Here, $\rho^{\mathrm{LCR1(or2)}} = V^{\mathrm{LCR1(or2)}}/V$ where the $V^{\mathrm{LCR1(or2)}}$ and V are the volumes of LCR1 (or LCR2) and MCR, respectively. $N_i^{\mathrm{LCR1(or2)}}$ and $N_i^{\mathrm{LCR1(or2)}}+1$ are the numbers of the same ion species as the created ion, i before and after a creation attempt in the LCR1 (or LCR2), $U_{N_i^{\mathrm{LCR1(or2)}}}$ and $U_{N_i^{\mathrm{LCR1(or2)}}+1}$ are the total interaction energies of ions before and after a creation attempt and q_i^{cre} is the charge of the created ion, i. V_0 is the applied voltage shown in Figure 1a where a voltage of V_0 is applied to the bottom and a zero voltage is applied to the top of the system. $\beta = 1/k_BT$ where k_B is the Boltzmann constant and T is the temperature. The constants B_i^{LCR1} and B_i^{LCR2} in Equations (1) are associated with the bulk ion concentration of an ion i at the LCR1 and LCR2 and are given by

$$B_i^{\mathrm{LCR1(or2)}} \equiv \beta\mu_i^{\mathrm{LCR1(or2)}} + \ln\frac{V}{\Lambda_i^3}. \tag{2}$$

Here, $\mu_i^{\mathrm{LCR1(or2)}}$ is the chemical potential of the created ion species i in LCR1 (or LCR2) and $\Lambda_i \equiv h/(2\pi m_i k_B T)^{1/2}$ where m_i is the mass of the created ion species. Unlike the original KLGCMC/MF approach where two different ion species share the same value of B_i, different values of B_i can be assigned to each ion

$$P_{N_i^{\mathrm{LCR1(or2)}} \to N_i^{\mathrm{LCR1(or2)}}-1} = \begin{cases} \min\left\{1, \dfrac{N_i^{\mathrm{LCR1}}}{\rho^{\mathrm{LCR1}}}\exp\left[-B_i^{\mathrm{LCR1}} - \beta\left(U_{N_i}{}^{\mathrm{LCR1}-1} - \beta\left(U_{N_i}{}^{\mathrm{LCR1}} + q_i^{\mathrm{del}}V_0\right)\right)\right], (\mathrm{LCR1}), \\ \min\left\{1, \dfrac{N_i^{\mathrm{LCR1}}}{\rho^{\mathrm{LCR1}}}\exp\left[-B_i^{\mathrm{LCR2}} - \beta\left(U_{N_i}{}^{\mathrm{LCR2}} + 1 - U_{N_i}{}^{\mathrm{LCR2}}\right)\right], \right\} \quad (\mathrm{LCR2}), \end{cases} \tag{3}$$

species in the current version of the eKLGCMC/MF simulation.

For ion deletion, an ion i in one of the two LCRs is randomly selected and an attempt to delete that ion is accepted or rejected with a probability of $P_{N_i^{\mathrm{LCR1(or2)}}\to N_i^{\mathrm{LCR1(or2)}}-1}$: where q_i^{del} is the charge of the deleted ion, i.

An ion move is attempted by sequentially selecting an ion in the MCR and moving that ion to one of the six nearest-neighbor sites. The move attempt is accepted or rejected with a probability of $P_{a\to b}$:

$$P_{a\to b} = \min\{1, \exp[-\beta(U_b - U_a)]\}, \tag{4}$$

where a and b are the ion positions before and after a move and U_a and U_b are the total interaction energy of ions before and after that move. In the current version of the simulation, one MC cycle is composed of move attempts for all the ions in the MCR and 20 creation and deletion trials for the ions in the LCRs.

Calculation of the total interaction energy of ions

The total interaction energy U_N for N ions in the model system is given as

$$\begin{aligned} U_N(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = &\sum_{i<j} u_{ij}(\mathbf{r}_i, \mathbf{r}_j) + \sum_i h(\mathbf{r}_i) \\ &+ \sum_i q_i \phi_{\mathrm{sf}}(\mathbf{r}_i) + \frac{1}{2}\sum_i q_i \phi_{\mathrm{rf}}(\mathbf{r}_i). \end{aligned} \tag{5}$$

Here, the first term in the right side of Equation (5) represents the direct interaction between ions, giving $u_{ij}(\mathbf{r}_i, \mathbf{r}_j)$ as

$$u_{ij}(\mathbf{r}_i, \mathbf{r}_j) = \frac{1}{4\pi\varepsilon_w}\frac{q_i q_j}{r_{ij}} + 4\varepsilon_{ij}^{\mathrm{LJ}}\left[\left(\frac{\sigma_{ij}^{\mathrm{LJ}}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}^{\mathrm{LJ}}}{r_{ij}}\right)^{6}\right], \tag{6}$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, ε_w is the water dielectric constant in bulk, q_i and q_j are the charges of ions i and j, and $\varepsilon_{ij}^{\mathrm{LJ}}$ and $\sigma_{ij}^{\mathrm{LJ}}$ are parameters associated with the Lennard-Jones (LJ) potential. The interaction energy $h(\mathbf{r}_i)$ in the second term of Equation (5) describes the hard–wall interaction of ion i with the inaccessible regions including the nanopore, membrane, and the MFRs and is given as

$$h(\mathbf{r}_i) = \begin{cases} \infty, & \mathbf{r}_i \in \text{in channel, membrane or MFRs,} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

The electrostatic interaction energy of ion i with fixed dipoles around the model nanopore in Figure 1a is accounted for by the third term of Equation (5) where the electrostatic potential due to the fixed dipoles is calculated using the modified PB equations.[16,21,32,33] The modified PB equations for the MCR, MFR1, and MFR2 are given as

$$\begin{cases} \nabla\cdot(\varepsilon(\mathbf{r})\nabla\phi_{sf}(\mathbf{r})) = -\rho_f(\mathbf{r}), & (\mathrm{MCR}), \\ \nabla\cdot(\varepsilon(\mathbf{r})\nabla\phi_{sf}(\mathbf{r})) - \varepsilon_w\kappa^2_{\mathrm{MFR1}}(\mathbf{r})(\phi_{sf}(\mathbf{r}) - V_0) = -\rho_f(\mathbf{r}), & (\mathrm{MFR1}), \\ \nabla\cdot(\varepsilon(\mathbf{r})\nabla\phi_{sf}(\mathbf{r})) - \varepsilon_w\kappa^2_{\mathrm{MFR2}}(\mathbf{r})\phi_{sf}(\mathbf{r}) = -\rho_f(\mathbf{r}), & (\mathrm{MFR2}). \end{cases} \tag{8}$$

Here, $\varepsilon(\mathbf{r})$ and $\kappa^2_{\mathrm{MFR1(or2)}}(\mathbf{r})$ are the space-dependent dielectric constant and the Debye–Hückel screening factor in MFR1 (or MFR2), respectively. The Debye–Hückel screening factor is defined as $\kappa^2_{\mathrm{MFR1(or2)}}(\mathbf{r}) \equiv \beta\sum_a z_a^2|e|^2 c_{a,0}^{\mathrm{MFR1(or2)}} H(\mathbf{r})/\varepsilon_w$ where $c_{a,0}^{\mathrm{MFR1(or2)}}$ is the bulk concentration of the ion species, a in the MFR1 (or MFR2) and $H(\mathbf{r}) = 1$ for ion accessible regions and otherwise $H(\mathbf{r}) = 0$. In Equation (8), $\rho_f(\mathbf{r})$ deals with the charge distribution of the fixed dipoles embedded in the membrane. The last term in the right side of Equation (5) describes the interaction of ion i and the reaction field potential, $\phi_{\mathrm{rf}}(\mathbf{r}_i)$ that is created by the induced charges on the nanopore/water, membrane/water, and MCR/MFR boundaries.[18,21,34–37] For our simulations, the reaction field potential $\phi_{\mathrm{rf}}(\mathbf{r}_i)$ is calculated using the basis set expansion (BSE) method where $\phi_{\mathrm{rf}}(\mathbf{r}_i)$ is expressed as a linear combination of the basis-set reaction field potentials $\phi_n^{\mathrm{rf}}(\mathbf{r})$,

$$\phi_{\mathrm{rf}}(\mathbf{r}) = \sum_n^M c_n \phi_n^{\mathrm{rf}}(\mathbf{r}). \tag{9}$$

Here, M is the number of basis functions and the expansion coefficient vector $\mathbf{C}^T = (c_1 c_2 \ldots c_M)$, which depends on the position and charge of ions, is expressed as

$$\mathbf{C} = \mathbf{S}^{-1}\mathbf{Q}, \tag{10}$$

where the elements of the $M \times M$ overlap matrix $\mathbf{S}$ of the basis functions and the generalized multipole moment vector $\mathbf{Q}$ are written as

$$S_{mn} = \int d\mathbf{r} b_m(\mathbf{r}) b_n(\mathbf{r}), \tag{11}$$

and

$$Q_n = \sum_i^N q_i b_n(\mathbf{r}_i), \tag{12}$$

respectively. Here, $b_n(\mathrm{r})$ is a basis function set that will be addressed in the next section and $\mathbf{r}_i$ is the position of ion i with a charge of q_i. The basis-set reaction field potential $\phi_n^{\mathrm{rf}}(\mathbf{r})$ in Equation (9) is given for a basis function $b_n(\mathbf{r})$ as

$$\phi_n^{\mathrm{rf}}(\mathbf{r}) = \phi_n^{\mathrm{inh}}(\mathbf{r}) - \phi_n^{\mathrm{blk}}(\mathbf{r}). \quad (13)$$

In Equation (13), $\phi_n^{\mathrm{inh}}(\mathbf{r})$ and $\phi_n^{\mathrm{blk}}(\mathbf{r})$ describe electrostatic potentials due to the basis function $b_n(\mathbf{r})$ in a system with inhomogeneous dielectric boundaries and in the homogeneous bulk water system, respectively, and can be acquired by solving

$$\nabla \cdot \left(\varepsilon(\mathbf{r})\nabla\phi_n^{\mathrm{inh}}(\mathbf{r})\right) - \varepsilon_w \kappa^2_{\mathrm{MFR1(or2)}}(\mathbf{r})\phi_n^{\mathrm{inh}}(\mathbf{r}) = -b_n(\mathbf{r}), \quad (14a)$$

$$\varepsilon_w \nabla^2 \phi_n^{\mathrm{blk}}(\mathbf{r}) = -b_n(\mathbf{r}). \quad (14b)$$

Here, the screening factor $\kappa_{\mathrm{MFR1(or2)}}$ is set to zero in MCR where ions are explicitly simulated.[18,21] Then the associated reaction field energy $U_{\mathrm{rf}} = 1/2\sum_i q_i\phi_{\mathrm{rf}}(\mathbf{r}_i)$ in Equation (5) can be expressed as

$$U_{\mathrm{rf}} = \frac{1}{2}\sum_n^M \sum_i^N c_n q_i \phi_n^{\mathrm{rf}}(\mathbf{r}_i). \quad (15)$$

The energy can be also written as

$$U_{\mathrm{rf}} = \frac{1}{2}\mathbf{Q}^T\widetilde{\mathbf{M}}\mathbf{Q}. \quad (16)$$

Here

$$\widetilde{\mathbf{M}} = \left(\mathbf{S}^{-1}\right)^T \mathbf{M}\mathbf{S}^{-1}, \quad (17)$$

where

$$M_{mn} = \int d\mathbf{r} b_m(\mathbf{r})\phi_n^{\mathrm{rf}}(\mathbf{r}). \quad (18)$$

Chebyshev polynomials as well as Legendre ones are tested as a basis set for the BSE method and comparisons with the exact reaction field energy values from the PB method are presented in Section "Results and Discussion."

Implementation of position-dependent diffusion coefficients of ions using the diffusion probability method

In the eKLGCMC/MF simulation method, position-dependent diffusion coefficients of ions are treated using the diffusion probability method that can not only handle the diffusivity of ions with different diffusion coefficients but also deal with the inhomogeneous diffusion coefficient of an ion species in a lattice-based simulation method.[31] As details of the diffusion probability method are introduced in Ref. 31, we describe the method briefly. In this method, a diffusion probability of ion i at the position of $\mathbf{r}_i$ is introduced as[31]

$$P_i^{\mathrm{dif}}(\mathbf{r}_i) \equiv \frac{D_i(\mathbf{r}_i)}{D_{\mathrm{ref}}}, \quad (19)$$

where D_{ref} is a reference diffusion coefficient and $D_{\mathrm{ref}} \geq D_i(\mathbf{r}_i)$ at any position of ion i. Then the total interaction energy in Equation (5) is slightly modified as

$$U_N(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = \sum_{i<j} u_{ij}(\mathbf{r}_i, \mathbf{r}_j) + \sum_i h(\mathbf{r}_i) + \sum_i q_i\phi_{\mathrm{sf}}(\mathbf{r}_i) + \frac{1}{2}\sum_i q_i\phi_{\mathrm{rf}}(\mathbf{r}_i) - k_B T \sum_i \ln P_i^{\mathrm{dif}}(\mathbf{r}_i), \quad (20)$$

where the last term in Equation (20) was added to account for the diffusion energy barrier.[38,39] It should be noted that Equations (19) and (20) are only applied to the ion move attempt and not applied to the ion creation or deletion attempt. For the eKLGCMC simulations, a position-dependent diffusion coefficient for ion i is modeled as

$$D_i(\mathbf{r}) = \begin{cases} D_i^{\mathrm{blk}} & (u \geq R_c \text{ and } |z| \geq L_{\mathrm{chn}}/2), \\ D_i^{\mathrm{chn}} + \left(D_i^{\mathrm{chn}} - D_i^{\mathrm{blk}}\right)g(u) & (0 < u < R_c \text{ and } |z| \geq L_{\mathrm{chn}}/2), \\ D_i^{\mathrm{chn}} & (|z| \leq L_{\mathrm{chn}}/2|), \end{cases} \quad (21)$$

where D_i^{blk} and D_i^{chn} are diffusion coefficients of ion i in bulk water and inside the channel, respectively, $u = \sqrt{x^2 + y^2 + (z| - L_{\mathrm{chn}}/2|)^2}$, $R_c = 10\text{Å}$, and $g(u) = 2(u/R_c)^3 - 3(u/R_c)^2$. We chose $R_c = 10\text{Å}$ because previous MD simulation studies revealed that the diffusion coefficients of ions begin to decrease nearly 10 Å away from the channel entrance of cyclic peptide nanotubes (CPNs) that possess a similar structure to the model nanopore in this study.[29,30] Then, the ion current for ion i can be evaluated as

$$I_i = \frac{6D_{\mathrm{ref}}\langle N_i^{\mathrm{cross}}\rangle}{\Delta L^2}, \quad (22)$$

where ΔL is the grid size or lattice spacing in the MCR and $\langle N_i^{\mathrm{cross}}\rangle$ is the average number of ion i crossing the cross-sectional area per time step.[18,31]

Description of the model nanopore system and details of the eKLGCMC/MF simulation

The model ion channel is composed of a model membrane, with the z axis taken to be normal to the membrane, and a small nanopore with radius 3.5 Å running through the membrane as shown in Figure 1a. In the membrane region 64 dipoles (a ring of eight dipoles

TABLE 1 Ion parameters

Ion	ε^{LJ} (kcal/Mol)[a,b]	$\sigma^{LJ}/2$ (Å)[a,b]	D^{blk} (Å²/ps)[c]	D^{chn} (Å²/ps)
Na^+	−0.0469	1.26	0.14	0.056
K^+	−0.0870	1.57	0.20	0.080
Cl^-	−0.150	2.02	0.20	0.080

[a]Taken from CHARMM27 force field.[42]
[b]$\varepsilon_{ij}^{LJ} = \sqrt{\varepsilon_{ii}^{LJ}\varepsilon_{jj}^{LJ}}$ and $\sigma_{ij}^{LJ} = \left(\sigma_{ii}^{LJ} + \sigma_{jj}^{LJ}\right)/2$.
[c]Taken from Ref. 43.

around the pore axis and eight rings along the axis) are placed in such a way that the negative ends of the dipoles are located closer to the nanopore, rendering the nanopore cation-selective. The length of one dipole is 5 Å and the charges of the positive and negative ends are $\pm 0.06\,|e|$, respectively. It has been reported that water molecules in a confined region like a small pore behave differently from bulk water,[40,41] but this issue is beyond the scope of this study and the water dielectric constant both in bulk and inside the pore is set to be $\varepsilon_w = 80\varepsilon_0$. The dielectric constant of the membrane region is taken to be $\varepsilon_m = 80\varepsilon_0$ or $\varepsilon_m = 2\varepsilon_0$ to assess the reaction field effect or dielectric self-energy effect due to the low dielectric membrane.[21,34–37] Ion-related parameters such as the LJ parameters and diffusion coefficients in bulk water and inside the nanopore are presented in Table 1 where the values of D_i^{chn} are set to be 0.4 D_i^{blk} for K^+, Na^+, and Cl^- on a basis of the molecular dynamics (MD) simulation results for the position-dependent diffusion coefficients of ions inside a cyclic peptide nanotube (CPN).[29,30] We used the ion radii from the Chemistry at Harvard Macromolecular Mechanics (CHARMM) force field instead of the hydrodynamic radii of ions because our previous MD simulations showed that the ions are partially dehydrated inside a small nanopore like the model system we used in this study.[10,42] A two-dimensional contour map and a one-dimensional profile for the relative position-dependent diffusion coefficient of ion i used in our simulations are illustrated in Figure 2. The B_i parameters in Equation (2) are associated with the bulk ion concentrations in the LCRs where the bulk ion concentrations are 0.25 M KCl and 0.05 M NaCl at the bottom and 0.05 M KCl and 0.1 NaCl at the top. In this work, three sets of B_i parameters are used for the bulk ion concentrations with no interionic interactions, with only van der Waals interactions between ions, and with full interionic interactions including Coulomb and van der Waals interactions, respectively. The B_i parameters corresponding to the ion concentrations in each condition are obtained using the iterative algorithm[44,45] (Figures S1–S3) and are presented in Table 2. The electrostatic focusing method is used to solve the electrostatic potentials due to the fixed dipoles under an applied voltage and the reaction field potentials. In the method electrostatic potentials for the whole system including the MFRs and MCR are evaluated on a grid of $61 \times 61 \times 81$ with a grid size of 1.0 Å, followed by a

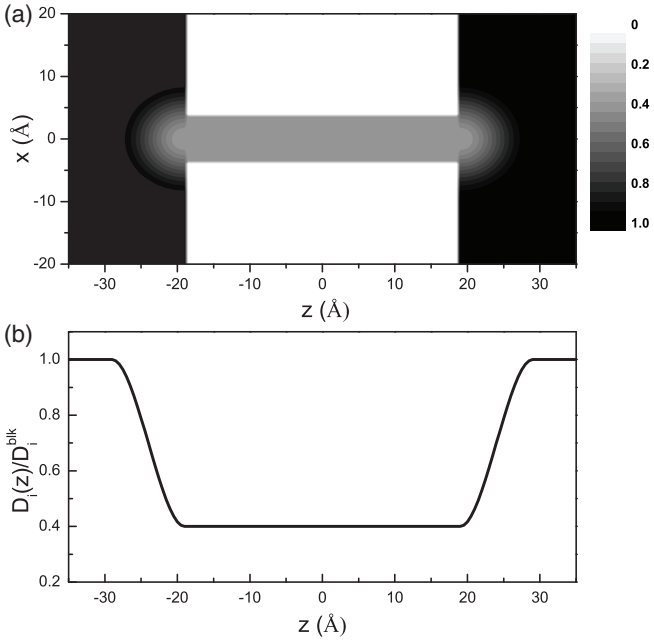


FIGURE 2 (a) Two-dimensional contour map and (b) a one-dimensional profile for the relative position-dependent diffusion coefficient of ion i ($i = K^+$, Na^+, and Cl^-). The diffusion coefficient of ions inside the channel, D_i^{chn} is set to be $D_i^{chn} = 0.4D_i^{blk}$

finer grid of $81 \times 81 \times 141$ with a grid size of 0.5 Å only in the MCR. Prior to starting eKLGCMC/MF simulations, the reaction field potentials for a total of 324 ($= 6 \times 6 \times 9$) basis functions are calculated and saved for later use. In the MCR the eKGLCMC simulations are performed with explicit ions on a $81 \times 81 \times 141$ grid with a grid size of 0.5 Å. Reflecting boundary conditions are used at the boundaries between the MFR and MCR regions in the x, y, and z directions. In this study, a grid size (ΔL) of 0.5 Å and a reference diffusion coefficient (D_{ref}) of 0.2 Å²/ps are used for the eKLGCMC/MF simulations, which lead to $\Delta t = 0.21$ ps.[31] For a given voltage a simulation is run over a total number of 1×10^9 MC cycles in the absence of the reaction field calculations or 2×10^8 MC cycles in the presence of the reaction field calculations, among which the first 5×10^7 MC cycles are reserved for equilibration and the remaining last MC cycles are used for data production.

Description of three-dimensional PNP and PB solvers

A 3D PNP solver that we have developed is used to acquire the ion current and concentration profiles for K^+, Na^+, and Cl^- as well as the electrostatic potentials for the model nanopore system in Figure 1a and to compare them to results from the eKLGCMC/MF simulations described above.[10] In the 3D PNP solver, the coupled PNP equations under a given applied voltage are numerically solved using a finite difference algorithm where the total

TABLE 2 Bulk ion concentrations and associated B_i parameters for the eKLGCMC/MF simulations

	Concentration (M)			B_i[a] (w/o ion-ion int.[b])			B_i (w/ ion-ion int.[c])		
System	K^+	Na^+	Cl^-	K^+	Na^+	Cl^-	K^+	Na^+	Cl^-
1	0.25	0.05	0.30	2.854 (2.879)[d]	1.249 (1.267)	3.041 (3.069)	2.487	0.841	2.632
2	0.05	0.10	0.15	1.250 (1.261)	1.940 (1.956)	2.346 (2.362)	0.942	1.616	2.020

[a]Defined in Equation (2).
[b]Neither electrostatic nor van der Waals interactions between ions considered in Equation (5).
[c]Both electrostatic and van der Waals interactions between ions considered in Equation (5).
[d]In parenthesis, B_i values in case of only van der Waals interactions between ions being considered.

electrostatic potentials and ion concentrations are simultaneously and self-consistently evaluated on a grid of $121 \times 121 \times 161$ with a grid size of 0.5 Å. In turn, the ion currents for each ion species can be calculated using the obtained electrostatic potentials and ion concentrations for a given voltage. A home-made 3D PB solver is also used to obtain exact values of the reaction field energy for a number of randomly generated ion configurations in the model nanopore system and to validate the BSE method by comparing the exact values to those approximated from the BSE method. For more details, refer to Ref. 10.

RESULTS AND DISCUSSION

Ion currents and concentration profiles in the absence of interionic interactions or in the presence of only short-range interactions between ions

In order to validate the eKLGCMC/MF simulation method for multiion systems, we switch off interionic interactions such as electrostatic and van der Waals interactions between ions to remove the effect of ion correlations on the ion transport through the model nanopore. This also means that the first term in Equation (5) is completely ignored while the other terms including the electrostatic interactions of ions with the fixed dipoles around the nanopore and with an applied voltage are still maintained. In addition, the dielectric constant of the membrane is set to be the same as that of water, that is, $\varepsilon_m = 80\varepsilon_0$ to avoid the reaction field effect as well. Because ion correlation vanishes and the reaction field effect is also ruled out, the eKLGCMC/MF results are expected to match well with the PNP calculations. Additional eKLGCMC/MF simulations in the presence of only van der Waals interactions between ions are also carried out to examine a weak ion correlation effect on ion translocation across the nanopore. In Figure 3, the I-V curves from the eKLGCMC/MF simulations indeed show remarkable agreement with those from the PNP solver in the absence of interionic interactions. The agreement between the two methods also demonstrates that the local control volume and diffusion probability methods

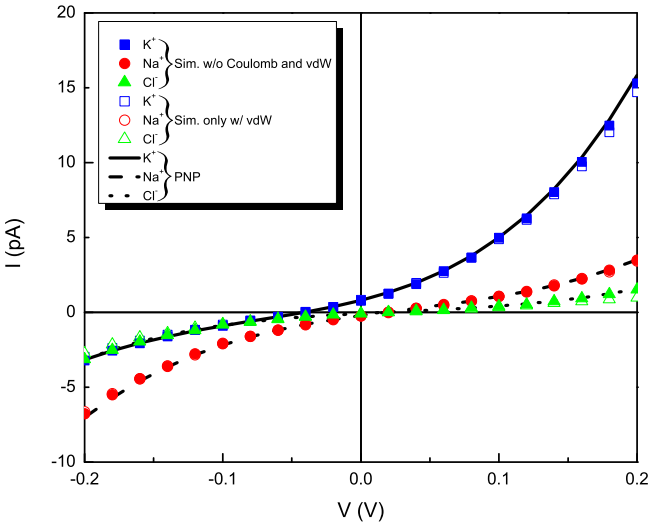


FIGURE 3 I-V curves from the eKLGCMC/MF simulations and their comparison with the PNP results in the absence of interionic interactions and in the presence of only van der Waals interactions between ions. The membrane dielectric constant is set to be $\varepsilon_m = 80\varepsilon_0$ and the bulk ion concentrations are 0.25 M KCl and 0.05 M NaCl at the bottom and 0.05 M KCl and 0.1 NaCl at the top. Inhomogeneous diffusion coefficients were used for ions as modeled in Equation (21) with parameters in Table 1

implemented in the eKLGCMC/MF simulation approach are well chosen for the transport of multiion species through a nanopore.

A little disagreement between the simulations and the PNP calculations at higher applied voltages is observed in Figure 3 when van der Waals interactions between ions are included in the simulations, indicating that ion correlations due to the short-range interionic interaction can affect the ion conduction through the nanopore. The I-V curves show asymmetric behavior due to asymmetric bulk ionic concentrations at the bottom and top LCRs, which also leads to small currents at $V_0 = 0$ V. Despite the highest bulk concentrations of Cl^-, the currents of Cl^- are less than the cation currents at high applied voltages, confirming that the model nanopore is designed as cation-selective. Excellent agreement between the eKLGCMC/MF simulations and PNP calculations is also found in Figure 4 where the concentration profiles for K^+, Na^+, and Cl^- at $V_0 = -0.2$ and 0.2 V from

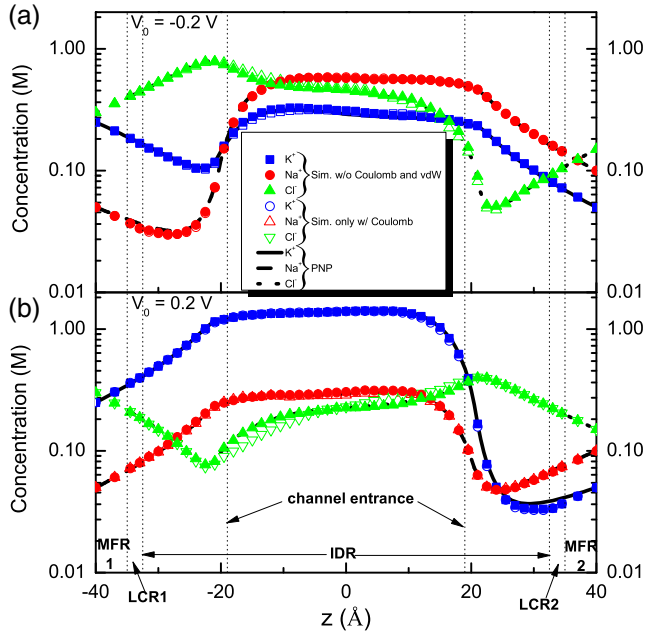


FIGURE 4 (a) and (b) comparisons between the eKLGCMC/MF simulations and PNP calculations for ion concentration profiles of K^+, Na^+, and Cl^- at $V_0 = -0.2$ V and $V_0 = 0.2$ V, respectively, in the absence of interionic interactions and in the presence of only van der Waals interactions between ions. The membrane dielectric constant and ion concentrations are the same as in Figure 3. The dotted lines indicate the boundaries between the MFRs and MCR at $z = \pm 35.0$ Å and between the LCRs and IDR at $z = \pm 32.5$ Å and the channel entrances at $z = \pm 19.0$ Å. note the semilogarithmic scale on the y axis

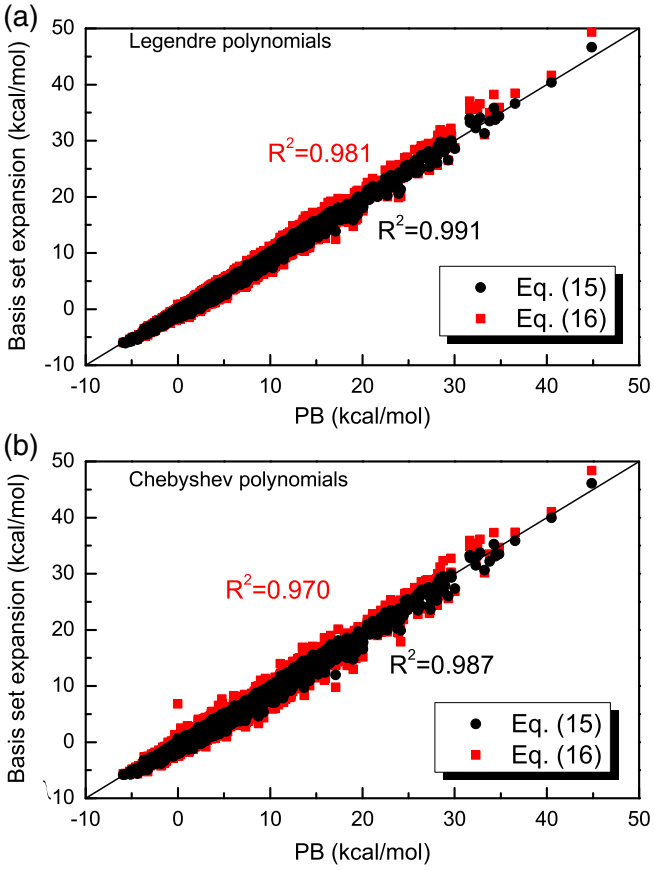


FIGURE 5 Comparisons of the reaction field energy for 5000 randomly generated configurations between the PB equation and the BSE approximation method based on (a) Legendre polynomials and (b) Chebyshev polynomials of the second kind. For calculations of the reaction field energy via the BSE method, both Equations (15) and (16) are used

the simulations are compared with those from the PNP solver in the absence of interionic interactions. As compared with those from the PNP calculations in the presence of the van der Waals interaction between ions, the Cl^- concentration profiles from the eKLGCMC/MF simulations show a small increase at the entrance of the nanopore and a small decrease at the exit of the pore in Figure 4, indicating slight Cl^- congestion due to the larger size of the anion.

Ion currents and ion concentration and electrostatic potential profiles in the presence of full interionic interactions

Prior to proceeding to the next systems with full interactions between ions, we validate the performance of the BSE method with which to treat the reaction field at the boundaries between the MFRs and the MCR as well as on the dielectric boundaries between the membrane and water interface by comparing the reaction field energies from the BSE method with exact values from the PB solver mentioned in Section "Results and Discussion." For the calculation of reaction field energies, we remove all the charges of the fixed dipoles around the pore, set the membrane dielectric constant, $\varepsilon_m = 2\varepsilon_0$ and generate a large number of configurations where K^+, Na^+, and Cl^- are randomly distributed in the MCR. In each configuration, at least one or two ions (mostly cations) are placed inside the pore in accordance with the assumption that the pore is cation-selective. The number of ions in the MCR varies from their mean values that are determined using the bulk concentrations of the three ions and the volume of the MCR. A total of 5000 configurations are generated and the reaction field energy for each configuration is evaluated using the PB solver and BSE approximation method. We test two different sets of polynomials, Legendre and Chebyshev polynomials, as a basis set to find a better set of polynomials for the BSE method in this work. Based on the Legendre polynomials $P_s(t)$, the basis function $b_n^{\mathrm{lgdr}}(\mathbf{r})$ is written as

$$b_n^{\mathrm{lgdr}}(\mathbf{r}) = H(\mathbf{r})\left[\frac{(2p+1)}{L_x}\frac{(2q+1)}{L_y}\frac{(2r+1)}{L_z}\right]^{1/2} P_p\left(\frac{2}{L_x}x\right)P_q\left(\frac{2}{L_y}y\right)P_r\left(\frac{2}{L_z}z\right), \quad (23)$$

and based on the Chebyshev polynomials of the second kind $U_s(t)$, $b_n^{\mathrm{cheb}}(\mathbf{r})$ is as

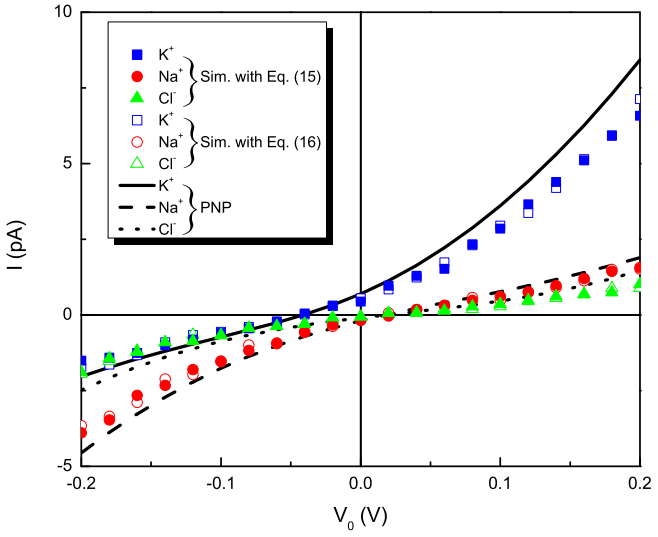


FIGURE 6 I-V curves from the eKLGCMC/MF simulations and their comparison with the PNP results in the presence of full interactions between ions. Dependence of I-V curves on the equations of the BSE method used for calculations of reaction field energy are also shown. The membrane dielectric constant is set to be $\varepsilon_m = 80\varepsilon_0$ and the bulk ion concentrations are 0.25 M KCl and 0.05 M NaCl at the bottom and 0.05 M KCl and 0.1 NaCl at the top. Inhomogeneous diffusion coefficients were used for ions as modeled in Equation (21) with parameters in Table 1

$$b_n^{\text{cheb}}(\mathbf{r}) = H(\mathbf{r})\left[\frac{4^3}{\pi^3 L_x L_y L_z}\right]^{1/2}\left[\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)\right]^{1/4} U_p\left(\frac{2}{L_x}x\right)U_q\left(\frac{2}{L_y}y\right)U_r\left(\frac{2}{L_z}z\right). \quad (24)$$

Here, p, q, and r are the degree of each polynomial, L_x, L_y, and L_z are the length of the MCR in each direction, and $H(\mathbf{r}) = 1$ for ion accessible regions and otherwise $H(\mathbf{r}) = 0$. Two related but different equations, Equations (15) and (16) in the BSE method are also independently employed to calculate the reaction field energies with a total of 324 ($= 6 \times 6 \times 9$) basis functions based on Equation (23) or (24).

Figure 5a,b (and Figure S4) demonstrates that both the Legendre and Chebyshev polynomials produce energy values comparable to those from the PB solver, but the Legendre polynomials make better evaluations for the reaction field energy than the Chebyshev ones. It is also shown in Figure 5 that Equation (15) gives better reaction field energies than Equation (16) presumably because larger integration errors are involved in

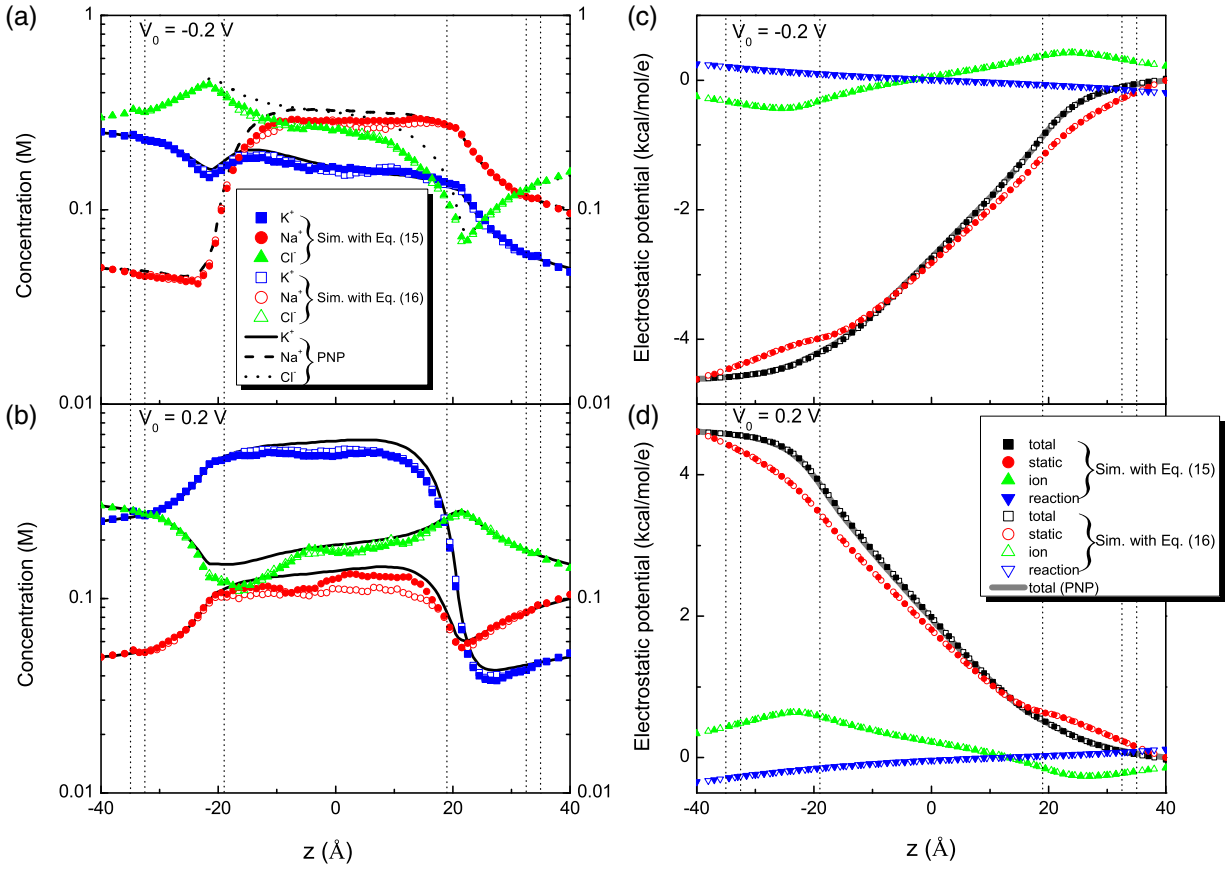


FIGURE 7 Comparisons between the eKLGCMC/MF simulations and PNP calculations for (a) and (b) ion concentration profiles and (c) and (d) electrostatic potential curves for K^+, Na^+, and Cl^- at $V_0 = -0.2$ V and $V_0 = 0.2$ V, respectively, in the presence of full interionic interactions. The membrane dielectric constant and ion concentrations are the same as in Figure 6. The dotted lines are the same as in Figure 4. Note the semilogarithmic scale on the y axis in the concentration profiles

Equation (16). Based on the results in Figure 5, the Legendre polynomials are selected as the basis functions for the BSE method in this study.

As a first example of the system with full interionic interactions, we choose a system with the dielectric constant of the membrane, $\varepsilon_m = 80\varepsilon_0$ to avoid the reaction field effect due to dielectric boundaries between the membrane/water interface so that we can assess only the ion correlation effect on the ion transport through the model nanopore. In Figure 6, the eKLGCMC/MF simulation results for the I-V curves show good agreement with the PNP results at lower voltages but deviate at higher voltages where the PNP solver overestimates the ion currents compared to the simulation outcomes. In consideration of the excellent agreement between the two methods in the absence of the interionic interaction in the previous subsection, the discrepancy between the eKLGCMC/MF simulation and the PNP solver at higher voltages in the presence of full interionic interactions is ascribed to the proper description of the ion correlation effect, which is lack in the PNP method. The I-V curves in Figure 6 indicate that the dependence of the ion currents on the choice of equations between Equations (15) and (16) are minimal in this system.

The ion concentration profiles at $V_0 = -0.2$ and 0.2 V in the system with $\varepsilon_m = 80\varepsilon_0$ under full interionic interactions are presented in Figure 7a,b where the concentration profiles from the eKLGCMC/MF simulations are in good agreement with the PNP results outside the nanopore. Inside the nanopore, however, the PNP calculations slightly overestimate ion concentrations compared to the eKLGCMC/MF simulations. At a high applied voltage of ± 0.2 V, the number of ions inside the model nanopore increases and as a result, ion correlations inside the pore also become stronger and this eventually leads to a discrepancy in the concentration profiles between the simulations and the PNP calculations. No distinct difference between Equations (15) and (16) is found in the concentration profiles in Figure 7a,b.

The total electrostatic potentials at $V_0 = -0.2$ and 0.2 V from the eKLGCMC/MF simulations and PNP solver are presented and compared in Figure 7c,d. The potential decomposition profiles of the total electrostatic potentials due to the three components of the system at $V_0 = -0.2$ and 0.2 V are also shown. The total electrostatic potentials from the two methods are nearly identical despite a small difference in the I-V curves and concentration profiles inside the nanopore. Although no reaction field potential is expected for the system with the same dielectric constant of membrane as that of water ($\varepsilon_m = \varepsilon_w = 80\varepsilon_0$), small reaction field potentials are observed at the boundaries between the MFRs and MCR due to interactions of explicit ions in the MCR with implicit ions described by $\kappa_{\mathrm{MFR1(or2)}}$ in MFRs.

In a second example of the system with full interionic interactions, the dielectric constant of membrane, ε_m is set to be $\varepsilon_m = 2\varepsilon_0$ that is close to that of a real membrane.

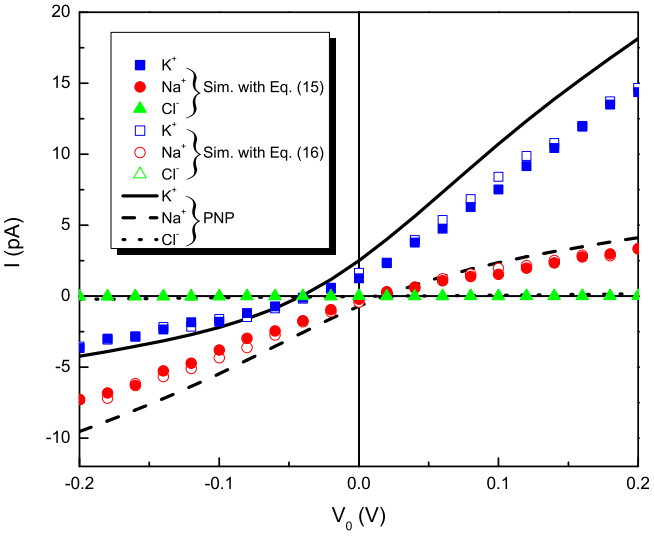


FIGURE 8 I-V curves from the eKLGCMC/MF simulations and their comparison with the PNP results in the presence of full interactions between ions. Dependence of I-V curves on the equations of the BSE method used for calculations of reaction field energy are also shown. The membrane dielectric constant is set to be $\varepsilon_m = 2\varepsilon_0$ and the bulk ion concentrations are 0.25 M KCl and 0.05 M NaCl at the bottom and 0.05 M KCl and 0.1 NaCl at the top. Inhomogeneous diffusion coefficients were used for ions as modeled in Equation (21) with parameters in Table 1

A much lower dielectric constant of the membrane induces a reaction field due to the dielectric difference between the membrane and water. Selectivity of the nanopore to cations is clearly found in the eKLGCMC/MF simulation and PNP results in Figure 8 where the cation conductances for K^+ and Na^+ are much larger than that of Cl^-. As compared to the eKLGCMC/MF simulation results, the PNP solver overestimates the ion currents over the whole range of applied voltages, which is obviously caused by ion correlation and reaction field effects which are not described properly in the PNP method. It has been reported that both the cation and anion currents are reduced by the reaction field effect.[36] The reduction of cation currents due to the reaction field is compensated for by the increase of cation currents owing to stronger electrostatic interactions between cations and the cation-selective nanopore through a much lower membrane dielectric constant. In the case of anion currents, however, the repulsive electrostatic interaction between anions and the cation-selective nanopore further diminishes the anion currents, and as a result, the anion currents become negligible as shown in Figure 8. The ion currents appear to have a small dependence on the choice of either Equation (15) or (16) in Figure 8.

Figure 9a,b shows ion concentration profiles for K^+, Na^+, and Cl^- at $V_0 = -0.2$ V and 0.2 V, respectively, in the system with $\varepsilon_m = 2\varepsilon_0$. These reveal a large discrepancy between the eKLGCMC/MF simulation results and PNP calculations. Ion selectivity in the model nanopore is also clearly observed in the ion concentration profiles in

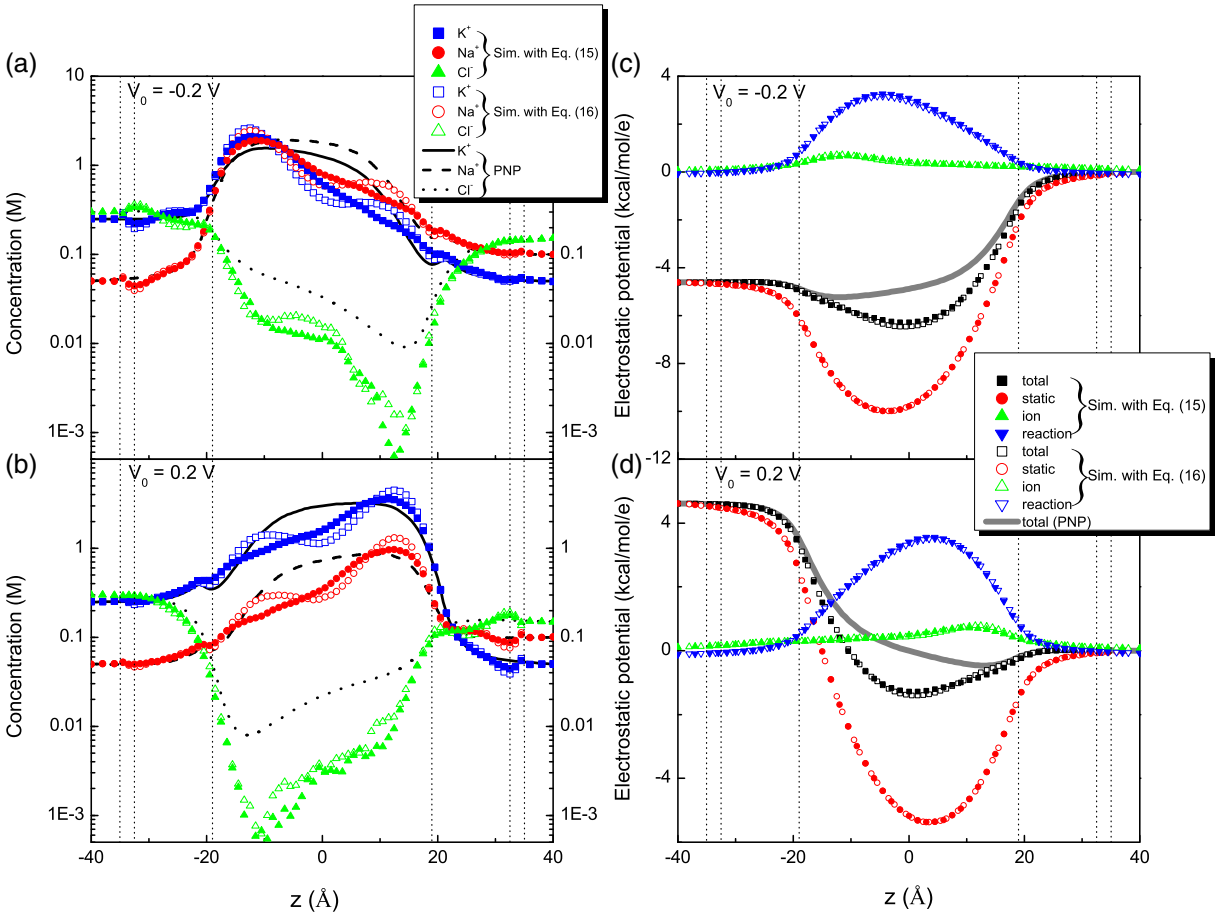


FIGURE 9 Comparisons between the eKLGCMC/MF simulations and PNP calculations for (a) and (b) ion concentration profiles and (c) and (d) electrostatic potential curves for K^+, Na^+, and Cl^- at $V_0 = -0.2$ V and $V_0 = 0.2$ V, respectively, in the presence of full interionic interactions. The membrane dielectric constant and ion concentrations are the same as in Figure 8. The dotted lines are the same as in Figure 4. Note the semilogarithmic scale on the *y* axis in the concentration profiles

Figure 9a,b where the ion concentrations for K^+ and Na^+ are much higher than that for Cl^-. Unlike the I-V curves in Figure 8, the ion concentrations inside the nanopore display a noticeable dependence on the choice of Equations (16) or (17) for the calculation of the reaction field energy. Because Figure 5 demonstrates that Equation (16) makes better evaluations for the reaction field energy than Equation (17), the ion concentration profiles from Equation (16) are expected to be more reliable. Comparisons between the eKLGCMC/MF simulations and PNP calculations for the total electrostatic potentials and the decomposition profiles of the total electrostatic potentials at $V_0 = -0.2$ V and 0.2 V are presented in Figure 9c,d. A large difference between the simulations and PNP results are also found in the total electrostatic potential profiles, where the eKLGCMC/MF simulations predict deeper potential wells inside the model nanopore. Decomposition profiles of the total electrostatic potentials from the eKLGCMC/MF simulations reveal that large reaction field potentials are created inside the nanopore that play a crucial role in the ion selectivity of the nanopore.

CONCLUDING REMARKS

In this study, an extension of the KLGCMC/MF (eKLGCMC/MF) simulation method was presented to assess the transport behavior of multicomponent ion mixtures through a cation-selective model nanopore. The diffusion probability algorithm, which was not included in the original KLGCMC/MF method, is implemented in the eKLGCMC/MF method to deal with the inhomogeneous diffusivity of ions. When the interionic interactions and reaction field effects are completely ignored, the simulation results show excellent agreement with the PNP calculations for the I-V curves and ion concentration profiles, validating the eKLGCMC/MF method. Comparison between the eKLGCMC/MF simulations and PNP calculations also demonstrates that inhomogeneous diffusion coefficients of ions are accurately described using the diffusion probability method in the eKLGCMC/MF method. In the system where the membrane dielectric constant is the same as that of water ($\varepsilon_m = \varepsilon_w = 80$) in the presence of full interionic interactions, the eKLGCMC/MF results predict

slightly smaller ion currents and concentrations than the PNP calculations do, revealing that ion correlations are important in a narrow nanopore. When the dielectric constant of the membrane ($\varepsilon_m = 2$) is much lower than that of water ($\varepsilon_w = 80$), the eKLGCMC/MF simulations generate considerably lower ion currents and concentrations inside the pore than the PNP calculations. This implies that the PNP theory overestimates the ion currents and concentrations inside the nanopore owing to insufficient description of both ion correlation and reaction field effects. Ion selectivity of the model nanopore is clearly illustrated in the I-V curves and ion concentration profiles from the low dielectric membrane system where the cation currents and concentrations inside the nanotube are much higher than the negligible anion currents and concentrations. Our study demonstrates that the proposed eKLGCMC/MF simulation approach can be utilized to elucidate unique phenomena occurring inside nanopores where multiple kinds of ions are involved. Although the BSE method accounts for the reaction field effect quite accurately, the method requires substantial computational time and cost and can be used only for systems with a small or moderate size. In order to avoid the computational cost for the reaction field calculation, we are currently developing a more efficient method for treatment of the reaction field effect that will be implemented in a next version of the eKLGCMC/MF simulation method.

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